

Introduction To RASPBERRY PI PICO By Making NOISE GENERATOR

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Raspberry Pi's first board based on a single microcontroller chip, the Raspberry Pi Pico (Fig. 1), is designed for physical computing—similar in concept to an Arduino—rather than perform the role of a general-purpose computer (like the others in the range). The Raspberry Pi Pico has 264kB RAM and 2MB flash memory, and it is programmable in MicroPython, CircuitPython, and C (https://en.wikipedia.org/wiki/Raspberry_Pi).

The purpose of this article is to introduce Raspberry Pi Pico (RPI

Pico), which operates at 133MHz with dual cores. Because it can run MicroPython, even a novice can start coding in a short amount of time. So, here is a project that can be done easily.

However, before jumping into the experiment/project, it would be a good idea to get acquainted with the RPi Pico microcontroller board. The following links would help you to ensure a great start:

- <https://projects.raspberrypi.org/en/projects/getting-started-with-the-pico>

- <https://www.electroschematics.com/raspberry-pi-pico-first-look-part-1/>

- <https://www.codrey.com/raspberry-pi/raspberry-pi-pico-quick-start/>

Firmware for Raspberry Pi Pico can be gathered from the following link:

<https://micropython.org/download/rp2-pico/>

You can download Thonny – Python IDE for beginners – from:

<https://thonny.org/>

The following link helps you learn how to program Raspberry Pi Pico using MicroPython:

<https://www.makeuseof.com/getting-started-micropython-raspberry-pi-pico/>

Making a noise generator

The Raspberry Pi Pico based noise generator project described here is in fact a simple tone generator setup realised with the help of a few external parts. Its schematic diagram is shown in Fig. 2.

As you can see, the setup delivers an analogue output through GP15 of Raspberry Pi Pico to drive a piezo speaker. The pushbutton switches S1 and S2 are wired to GP0 and GP1 to control the tone generation.

The element used here for sound is not a common piezo buzzer; it is a piezo speaker that is also called 'motherboard speaker.' This piezo speaker (Fig. 3) is usually employed to obtain beep sound from computer



Fig. 1: Raspberry Pi Pico board

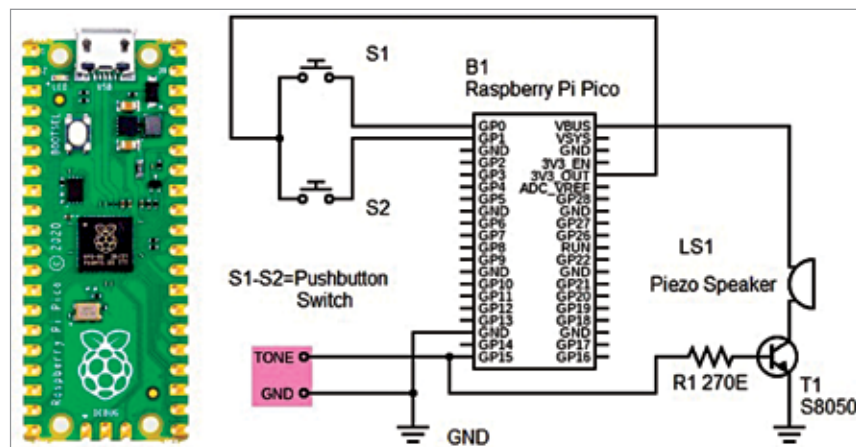


Fig. 2: Circuit diagram

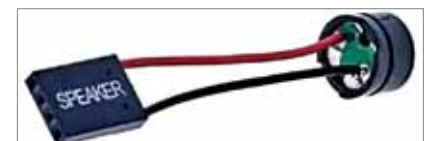


Fig. 3: Piezo speaker